

MATHCLEAR · VTU 2021 SCHEME

21MAT11 Complete Revision Handbook

Calculus and Differential Equations

All module questions, full solutions, formula explanations, mark allocation and plain-English intuition.

60+ raw-mark preparation target

Preparation support—not an examination guarantee.

Preparation target: 60+ raw marks. Study above the bare minimum. No question set can guarantee a result.

Reliable method: Read the intuition → cover the solution → recall the formula → write the answer → compare and correct.

Source policy: Official VTU 2021 Scheme syllabus and model papers are the primary authority. Supplied VTU papers provide recurrence evidence. TIE/SIMP and other internet material are secondary cross-checks only where topics match the official syllabus.

VTU answer order: Given/definition → formula/theorem → substitution → stepwise working → boxed final answer.

Essential Full Forms

VTU: Visvesvaraya Technological University

SEE: Semester End Examination

CIE: Continuous Internal Evaluation

ODE: Ordinary Differential Equation

PDE: Partial Differential Equation

CF: Complementary Function

PI: Particular Integral

RK4: Fourth-Order Runge-Kutta Method

LHS / RHS: Left-Hand Side / Right-Hand Side

21MAT11 – Calculus and Differential Equations

Independently Authored VTU 2021-Scheme Practice Paper with Solutions

Time: 3 hours

Maximum marks: 100

Instructions

Answer **five full questions**, choosing **one full question from each module**.

Each full question carries **20 marks**.

The observed VTU part-mark pattern **6 + 7 + 7** is followed.

Both alternatives are solved here for practice.

Questions are independently authored using recurring, beginner-solvable patterns found in:

M1_ALL_OCR.txt

M1_M2_OFFICIAL_EXTRACT.txt

Module 1 – Differential Calculus I

Q1 – 20 marks

Q1(a) – 6 marks

Prompt: With the usual polar–curve notation, prove that

$$\tan \phi = r \frac{d\theta}{dr},$$

where ϕ is the angle between the radius vector and the tangent.

Intuition: Resolve a small displacement along the curve into radial and perpendicular components. Their ratio gives the tangent of the angle.

Stepwise solution and marking

Let $P(r, \theta)$ and $Q(r + dr, \theta + d\theta)$ be neighboring points. [1 mark]

The displacement in the radial direction is dr . [1 mark]

The displacement perpendicular to the radius vector is $r, d\theta$. [1 mark]

Hence, for the angle ϕ between the tangent and radius vector,

$$\tan \phi = \frac{\text{perpendicular displacement}}{\text{radial displacement}}.$$

[1 mark]

Therefore,

$$\tan \phi = \frac{r, d\theta}{dr}.$$

[1 mark]

Thus,

$$\boxed{\tan \phi = r \frac{d\theta}{dr} = \frac{r}{dr/d\theta}}.$$

[1 mark]

Final answer:

$$\boxed{\tan \phi = r \frac{d\theta}{dr}}.$$

Q1(b) – 7 marks

Prompt: Find the angle of intersection of the curves

$$r = 2a \sin \theta, \quad r = 2a \cos \theta.$$

Intuition: First find the common point, then calculate the angle each tangent makes with the common radius vector.

Stepwise solution and marking

At a non–polar intersection,

$$2a \sin \theta = 2a \cos \theta,$$

so

$$\tan \theta = 1, \quad \theta = \frac{\pi}{4}.$$

[1 mark]

For $r_1 = 2a \sin \theta$,

$$\left[\frac{dr_1}{d\theta} = 2a \cos \theta. \right]$$

[1 mark]

Using $\tan \phi = r / (dr/d\theta)$,

$$\left[\tan \phi_1 = \frac{2a \sin \theta}{2a \cos \theta} = \tan \theta. \right]$$

At $\theta = \pi/4$,

$$\left[\phi_1 = \frac{\pi}{4}. \right]$$

[1 mark]

For $r_2 = 2a \cos \theta$,

$$\left[\frac{dr_2}{d\theta} = -2a \sin \theta. \right]$$

[1 mark]

Therefore,

$$\left[\tan \phi_2 = \frac{2a \cos \theta}{-2a \sin \theta} = -\cot \theta. \right]$$

At $\theta = \pi/4$,

$$\left[\tan \phi_2 = -1, \text{quad } \phi_2 = -\frac{\pi}{4}. \right]$$

[1 mark]

The angle between the curves is

$$\left[\alpha = |\phi_1 - \phi_2| = \left| \frac{\pi}{4} + \frac{\pi}{4} \right| = \frac{\pi}{2}. \right]$$

[1 mark]

Hence the curves intersect orthogonally. **[1 mark]**

Final answer:

$$\left[\boxed{\alpha = \frac{\pi}{2} = 90^\circ}. \right]$$

Q1© €” 7 marks

Prompt: Find the radius of curvature of

$$\left[y = x^2 \right]$$

at the point $(1, 1)$.

Intuition: For a Cartesian curve, the radius of curvature depends on the first and second derivatives.

Stepwise solution and marking

The Cartesian radius-of-curvature formula is

$$\left[\rho = \frac{\left(1 + (y')^2 \right)^{3/2}}{|y''|}. \right]$$

[1 mark]

Given $y = x^2$,

$$\left[y' = 2x. \right]$$

[1 mark]

Also,

$$\left[y'' = 2. \right]$$

[1 mark]

At $x = 1$,

$$\left[y' = 2, \text{quad } y'' = 2. \right]$$

[1 mark]

Substitute:

$$\left[\rho = \frac{(1+2^2)^{3/2}}{2} \right]$$

[1 mark]

Therefore,

$$\left[\rho = \frac{5^{3/2}}{2} = \frac{5\sqrt{5}}{2} \right]$$

[1 mark]

The radius is positive, as required. **[1 mark]**

Final answer:

$$\left[\boxed{\rho = \frac{5\sqrt{5}}{2}} \right]$$

OR

Q2 € 20 marks

Q2(a) € 6 marks

Prompt: If p is the perpendicular distance from the pole to the tangent of a polar curve, prove that

$\left[\frac{1}{p^2} \right]$

$$\frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2.$$

Intuition: The perpendicular p is related to the radius vector through the right triangle formed with the tangent.

Stepwise solution and marking

Let ϕ be the angle between the radius vector and tangent. Then

$$[p = r \sin \phi.]$$

[1 mark]

Therefore,

$$\left[\frac{1}{p^2} = \frac{1}{r^2 \sin^2 \phi} \right]$$

[1 mark]

Since

$$\left[\csc^2 \phi = 1 + \cot^2 \phi, \right]$$

we obtain

$$\left[\frac{1}{p^2} = \frac{1}{r^2} (1 + \cot^2 \phi) \right]$$

[1 mark]

The polar tangent formula gives

$$\left[\tan \phi = \frac{r}{dr/d\theta} \right]$$

Hence,

$$\left[\cot \phi = \frac{1}{r} \frac{dr}{d\theta} \right]$$

[1 mark]

Substitution gives

$$\left[\frac{1}{p^2} = \frac{1}{r^2} \left[1 + \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 \right] \right]$$

[1 mark]

Therefore,

$$\boxed{\frac{1}{p^2}}$$

$$\frac{1}{r^2}$$

$$\frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2 .$$

[1 mark]

Q2(b) € 7 marks

Prompt: Find the pedal equation of the logarithmic spiral

$$[r = ae^{\theta} .]$$

Intuition: Differentiate the polar equation and substitute into the standard relation involving p and r .

Stepwise solution and marking

Given

$$[r = ae^{\theta} .]$$

[1 mark]

Differentiate:

$$\left[\frac{dr}{d\theta} = ae^{\theta} = r . \right]$$

[1 mark]

Use

$$\left[\frac{1}{r^2} \right]$$

$$\frac{1}{r^2}$$

$$\frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2.$$

[1 mark]

Substituting $dr/d\theta = r$ gives

$$\left[\frac{1}{p^2} \right]$$

$$\frac{1}{r^2} + \frac{r^2}{r^4}.$$

[1 mark]

Thus,

$$\left[\frac{1}{p^2} = \frac{2}{r^2} \right]$$

[1 mark]

Therefore,

$$\left[p^2 = \frac{r^2}{2} \right]$$

[1 mark]

Since p is a distance,

$$\left[\boxed{p = \frac{r}{\sqrt{2}}} \right]$$

[1 mark]

Q2 © 7 marks

Prompt: Find the radius of curvature of the cardioid

$$\left[r = a(1 + \cos \theta) \right]$$

at $\theta = 0$.

Intuition: Use the polar radius-of-curvature formula and evaluate the derivatives at the specified point.

Stepwise solution and marking

The polar formula is

$$\left[\rho = \frac{\left(r^2 + (r')^2 \right)^{3/2}}{|r^2 + 2(r')^2 - r r''|} \right]$$

[1 mark]

Given

$$\left[r = a(1 + \cos \theta) \right]$$

Therefore,

$$\left[r' = -a \sin \theta \right]$$

[1 mark]

Also,

$$\left[r'' = -a \cos \theta \right]$$

[1 mark]

At $\theta = 0$,

$$\left[r = 2a, \quad r' = 0, \quad r'' = -a \right]$$

[1 mark]

Numerator:

$$\left[\left(r^2 + (r')^2 \right)^{3/2} = \left(4a^2 \right)^{3/2} = 8a^3 \right]$$

[1 mark]

Denominator:

$$\left[r^2 + 2(r')^2 - r r'' = 4a^2 - 2a(-a) = 6a^2 \right]$$

[1 mark]

Hence,

$$\left[\rho = \frac{8a^3}{6a^2} = \boxed{\frac{4a}{3}}. \right]$$

[1 mark]

Module 1 Formula Glossary

r : radius vector.

θ : polar angle.

ϕ : angle between radius vector and tangent.

p : perpendicular distance from the pole to tangent.

ρ : radius of curvature.

Cartesian radius of curvature:

$$\left[\rho = \frac{[1+(y')^2]^{3/2}}{|y''|}. \right]$$

Polar radius of curvature:

$$\left[\rho = \frac{[r^2+(r')^2]^{3/2}}{|r^2+2(r')^2-r r''|}. \right]$$

Pedal relation:

[\frac{1}{p^2}

$$\frac{1}{r^2}$$
$$\frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2.$$

Module 2 – Differential Calculus II

Q3 – 20 marks

Q3(a) – 6 marks

Prompt: Expand $e^x \cos x$ by Maclaurin's series up to the term containing x^4 .

Intuition: Write the known series for e^x and $\cos x$, then multiply and collect like powers.

Stepwise solution and marking

Use

$$[e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots]$$

[1 mark]

Also,

$$[\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots]$$

[1 mark]

Multiply:

$$[e^x \cos x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} \right) \left(1 - \frac{x^2}{2} + \frac{x^4}{24} \right)]$$

[1 mark]

Constant, linear and quadratic terms are

$$[1 + x + \left(\frac{1}{2} - \frac{1}{2} \right) x^2 = 1 + x.]$$

[1 mark]

Cubic coefficient:

$$[\frac{1}{6} - \frac{1}{2} = -\frac{1}{3}.]$$

Quartic coefficient:

$$[\frac{1}{24} - \frac{1}{4} + \frac{1}{24} = -\frac{1}{6}.]$$

[1 mark]

Therefore,

$$[\boxed{e^x \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + O(x^5)}.]$$

[1 mark]

Q3(b) – 7 marks

Prompt: If

$$[u = x^2 + y^2, \quad x = t^2, \quad y = 2t,]$$

find du/dt using partial differentiation.

Intuition: Since both x and y depend on t , apply the total derivative chain rule.

Stepwise solution and marking

The total derivative formula is

[\frac{du}{dt}]

$$\frac{\partial u}{\partial x} \frac{dx}{dt}$$
$$\frac{\partial u}{\partial y} \frac{dy}{dt}.]$$

[1 mark]

Since $u = x^2 + y^2$,

$$\frac{\partial u}{\partial x} = 2x.]$$

[1 mark]

Similarly,

$$\frac{\partial u}{\partial y} = 2y.]$$

[1 mark]

From $x = t^2$,

$$\frac{dx}{dt} = 2t.]$$

[1 mark]

From $y = 2t$,

$$\frac{dy}{dt} = 2.]$$

[1 mark]

Therefore,

$$\frac{du}{dt} = (2x)(2t) + (2y)(2).]$$

Substituting $x = t^2$ and $y = 2t$,

$$\frac{du}{dt} = 4t^3 + 8t.]$$

[1 mark]

Hence,

$$\boxed{\frac{du}{dt} = 4t^3 + 8t}.]$$

[1 mark]

Q3 © " 7 marks

Prompt: Examine

$$f(x,y) = x^2 + y^2 - 4x + 6y + 20]$$

for an extreme value.

Intuition: Locate the stationary point and use second derivatives to classify it.

Stepwise solution and marking

Compute

$$f_x = 2x - 4, \quad f_y = 2y + 6.]$$

[1 mark]

Set $f_x = f_y = 0$:

$$2x - 4 = 0, \quad 2y + 6 = 0.]$$

Hence,

$$(x,y) = (2,-3).]$$

[1 mark]

Second derivatives are

$$[f_{xx}=2, \quad f_{yy}=2, \quad f_{xy}=0.]$$

[1 mark]

The discriminant is

$$[D=f_{xx}f_{yy}-(f_{xy})^2=2(2)-0=4.]$$

[1 mark]

Since $D > 0$ and $f_{xx} > 0$, the point is a minimum. **[1 mark]**

Its value is

$$[f(2,-3)=4+9-8-18+20=7.]$$

[1 mark]

Therefore, the minimum value is

$$[\boxed{7} \text{ at } (2,-3).]$$

[1 mark]

OR

Q4 € 20 marks

Q4(a) € 6 marks

Prompt: Evaluate:

$$[\text{(i)} \quad \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}, \quad \text{(ii)} \quad \lim_{x \rightarrow 0} (1+x)^{1/x}.]$$

(i) Solution € 3 marks

Let the limit be L . Then

[$\log L$

$$\lim_{x \rightarrow 0} \frac{\log(\sin x/x)}{x^2}.$$

[1 mark]

Using

$$\sin x = 1 - \frac{x^2}{6} + O(x^4),$$

we get

$$\log\left(\frac{\sin x}{x}\right) = -\frac{x^2}{6} + O(x^4).$$

Therefore,

$$\log L = -\frac{1}{6}.$$

[1 mark]

Hence,

$$\boxed{L = e^{-1/6}}.$$

[1 mark]

(ii) Solution € 3 marks

Let the limit be M . Then

[\log M

$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x}.$

[1 mark]

By L'Hôpital's rule,

[\log M

$$\lim_{x \rightarrow 0} \frac{1}{(1+x)} = 1.$$

[1 mark]

Therefore,

$$\boxed{M=e}.$$

[1 mark]

Final answers:

$$\boxed{\text{(i)} \ e^{-1/6}, \text{quad} \ \text{(ii)} \ e}.$$

Q4(b) € 7 marks

Prompt: If

$$u = x^2y + yz, \text{quad} \ x = t, \text{quad} \ y = t^2, \text{quad} \ z = t^3,$$

find du/dt using the total derivative.

Intuition: Add the effect of changing each independent variable.

Stepwise solution and marking

Use

$$\frac{du}{dt} = u_x \frac{dx}{dt} + u_y \frac{dy}{dt} + u_z \frac{dz}{dt}.$$

[1 mark]

The partial derivatives are

$$u_x = 2xy.$$

[1 mark]

Also,

$$u_y = x^2 + z, \text{quad} \ u_z = y.$$

[1 mark]

The derivatives of the substitutions are

$$\frac{dx}{dt} = 1, \text{quad} \ \frac{dy}{dt} = 2t, \text{quad} \ \frac{dz}{dt} = 3t^2.$$

[1 mark]

Therefore,

$$\frac{du}{dt} = (2xy)(1) + (x^2 + z)(2t) + y(3t^2).$$

[1 mark]

Substitute $x = t, y = t^2, z = t^3$:

$$\frac{du}{dt} = 2t^3 + 2t(t^2 + t^3) + 3t^4.$$

[1 mark]

Thus,

$$\boxed{\frac{du}{dt} = 4t^3 + 5t^4}.$$

[1 mark]

A direct check gives $u = t^4 + t^5$, whose derivative is also $4t^3 + 5t^4$.

Q4 © €” 7 marks

Prompt: If

$$[u=x+y+z, \quad v=y+z, \quad w=z+x,]$$

find

$$[\frac{\partial(u,v,w)}{\partial(x,y,z)}.]$$

Intuition: Form the matrix of first partial derivatives and evaluate its determinant.

Stepwise solution and marking

By definition,

[\frac{\partial(u,v,w)}{\partial(x,y,z)}

|u_x u_y u_z v_x v_y v_z w_x w_y w_z|

.]

[1 mark]

For $u = x + y + z$,
[(u_x,u_y,u_z)=(1,1,1).]

[1 mark]

For $v = y + z$,
[(v_x,v_y,v_z)=(0,1,1).]

[1 mark]

For $w = z + x$,
[(w_x,w_y,w_z)=(1,0,1).]

[1 mark]

Therefore,
[J=

|1 1 1 0 1 1 0 1|

.]

[1 mark]

Expanding along the first row,
[J=1(1)-1(-1)+1(-1)=1.]

[1 mark]

Hence,
[\boxed{\frac{\partial(u,v,w)}{\partial(x,y,z)}=1}.]

[1 mark]

Module 2 Formula Glossary

Maclaurin series:
[f(x)=f(O)+xf'(O)+\frac{x^2}{2!}f''(O)+\cdots.]
Total derivative:

[\frac{du}{dt}

$u_x \frac{dx}{dt} + u_y \frac{dy}{dt} + u_z \frac{dz}{dt}.$]

Jacobian:

$\left[\frac{\partial(u,v,w)}{\partial(x,y,z)} \right]$

$\det\left[\frac{\partial(u,v,w)}{\partial(x,y,z)}\right]$.

Two-variable extreme-value test:

$D = f_{xx}f_{yy} - (f_{xy})^2$.

$D > 0, f_{xx} > 0$: minimum.

$D > 0, f_{xx} < 0$: maximum.

$D < 0$: saddle point.

L'Hôpital's rule: For a $0/0$ or ∞/∞ form,

[\lim\frac{f(x)}{g(x)}

$\lim\frac{f'(x)}{g'(x)}$]

when the required conditions hold.

Module 3 – First-Order ODEs

Q5 – 20 marks

Q5(a) – 6 marks

Prompt: Solve the Bernoulli differential equation

$$\left[\frac{dy}{dx} + y = xy^2. \right]$$

Intuition: Because the equation contains y^2 , use the substitution $v = y^{-1}$ to convert it into a linear equation.

Stepwise solution and marking

The equation is Bernoulli's equation with $n = 2$. Divide by y^2 :

$$\left[y^{-2}y' + y^{-1} = x. \right]$$

[1 mark]

Put

$$\left[v = y^{-1}. \right]$$

Then

$$\left[v' = -y^{-2}y'. \right]$$

[1 mark]

Therefore,

$$\left[-v' + v = x, \right]$$

or

$$\left[v' - v = -x. \right]$$

[1 mark]

Its integrating factor is

$$\left[IF = e^{\int -1, dx} = e^{-x}. \right]$$

[1 mark]

Hence,

$$\left[\frac{d}{dx}(ve^{-x}) = -xe^{-x}. \right]$$

Since

$$\left[\int -xe^{-x}, dx = (x+1)e^{-x}, \right]$$

$$\left[ve^{-x} = (x+1)e^{-x} + C. \right]$$

[1 mark]

Thus,

$$\left[v = x+1 + Ce^x. \right]$$

Since $v = 1/y$,

$$\left[\boxed{y = \frac{1}{x+1+Ce^x}}. \right]$$

Also, $y = 0$ is a solution. **[1 mark]**

Q5(b) € 7 marks

Prompt: Find the orthogonal trajectories of

$$[x^2 + y^2 = 2cx,]$$

where c is the parameter.

Intuition: Eliminate the parameter to obtain the slope of the given family, then replace it by its negative reciprocal.

Stepwise solution and marking

Differentiate:

$$[2x + 2y \frac{dy}{dx} = 2c.]$$

[1 mark]

From the original family,

$$[c = \frac{x^2 + y^2}{2x}.]$$

[1 mark]

Substitute:

$$[x + y \frac{dy}{dx} = \frac{x^2 + y^2}{x}.]$$

[1 mark]

Therefore,

$$[y \frac{dy}{dx} = \frac{y^2}{x}, \quad \frac{dy}{dx} = \frac{y}{x}.]$$

[1 mark]

The slope of the orthogonal trajectories is the negative reciprocal:

$$[\frac{dx}{dy} = -\frac{x}{y}.]$$

[1 mark]

Hence,

$$[y, dy = -x, dx.]$$

Integrating,

$$[\frac{y^2}{2} = -\frac{x^2}{2} + C.]$$

[1 mark]

Therefore,

$$[\boxed{x^2 + y^2 = C}.]$$

[1 mark]

Q5© € 7 marks

Prompt: Solve

$$[p^2 - (x+y)p + xy = 0, \quad p = \frac{dy}{dx}.]$$

Intuition: Factor the equation as an algebraic equation in p , then solve the resulting first-order equations separately.

Stepwise solution and marking

Factor:

$$[p^2 - (x+y)p + xy = (p-x)(p-y).]$$

[1 mark]

Therefore,

$$[(p-x)(p-y) = 0.]$$

[1 mark]

First case:

$$[p = x \quad \rightarrow \quad \frac{dy}{dx} = x.]$$

[1 mark]

Integrating,

$$[y = \frac{x^2}{2} + C_1.]$$

[1 mark]

Second case:

$$[p = y \quad \rightarrow \quad \frac{dy}{dx} = y.]$$

[1 mark]

Hence,

$$[\frac{dy}{y} = dx \quad \rightarrow \quad \log|y| = x + C,]$$

so

$$[y = C_2 e^x.]$$

[1 mark]

The two solution families are

$$[\boxed{y = \frac{x^2}{2} + C_1} \quad \text{or} \quad \boxed{y = C_2 e^x}.]$$

[1 mark]

OR

Q6 € 20 marks

Q6(a) € 6 marks

Prompt: Solve

$$[(2x+y), dx + (x+2y), dy = 0.]$$

Intuition: Check exactness and integrate one coefficient.

Stepwise solution and marking

Let

$$[M = 2x + y, \quad N = x + 2y.]$$

[1 mark]

Check:

$$[M_y = 1, \quad N_x = 1.]$$

Hence the equation is exact. **[1 mark]**

Let $F_x = M$. Then

$$[F = \int (2x+y) dx = x^2 + xy + \phi(y).]$$

[1 mark]

Differentiate with respect to y :

$$[F_y = x + \phi'(y).]$$

[1 mark]

Equate this to $N = x + 2y$:

$$[\phi'(y) = 2y, \quad \phi(y) = y^2.]$$

[1 mark]

Therefore,

$$[\boxed{x^2+xy+y^2=C}.]$$

[1 mark]

Q6(b) – 7 marks

Prompt: A metal body is initially at 90°C and is placed in a room maintained at 30°C . After 10 minutes its temperature is 60°C . Find its temperature after 20 minutes.

Intuition: Newton's law states that the excess temperature above the surroundings decreases exponentially.

Stepwise solution and marking

Newton's law is

$$[\frac{dT}{dt}=-k(T-30).]$$

[1 mark]

Its solution is

$$[T-30=Ce^{-kt}.]$$

[1 mark]

At $t = 0$, $T = 90$, so

$$[90-30=C, \quad C=60.]$$

Hence,

$$[T-30=60e^{-kt}.]$$

[1 mark]

At $t = 10$, $T = 60$:

$$[30=60e^{-10k}.]$$

Thus,

$$[e^{-10k}=\frac{1}{2}.]$$

[1 mark]

At $t = 20$,

$$[e^{-20k}=(e^{-10k})^2=\left(\frac{1}{2}\right)^2=\frac{1}{4}.]$$

[1 mark]

Therefore,

$$[T-30=60\left(\frac{1}{4}\right)=15.]$$

[1 mark]

Hence,

$$[\boxed{T=45^\circ\text{C}}.]$$

[1 mark]

Q6(c) – 7 marks

Prompt: Find the general and singular solutions of Clairaut's equation

$$[y=px+p^2, \quad p=\frac{dy}{dx}.]$$

Intuition: In Clairaut's equation, replace the constant slope p by a parameter for the general solution. The singular solution is the envelope of that family.

Stepwise solution and marking

The standard Clairaut form is

$$[y = px + f(p).]$$

Here,

$$[f(p) = p^2.]$$

[1 mark]

Differentiate:

$$[p = p + (x + 2p) \frac{dp}{dx}.]$$

[1 mark]

Thus,

$$[(x + 2p) \frac{dp}{dx} = 0.]$$

[1 mark]

From $dp/dx = 0$, let $p = C$. Then

$$[\boxed{y = Cx + C^2}]$$

is the general solution. **[1 mark]**

For the singular solution,

$$[x + 2p = 0, \quad p = -\frac{x}{2}.]$$

[1 mark]

Substitute into $y = px + p^2$:

$$[y = -\frac{x^2}{2} + \frac{x^2}{4}.]$$

[1 mark]

Hence the singular solution is

$$[\boxed{y = -\frac{x^2}{4}}.]$$

[1 mark]

Module 3 Formula Glossary

ODE: Ordinary Differential Equation.

p : shorthand for dy/dx .

Linear first-order ODE:

$$[y' + P(x)y = Q(x).]$$

IF: Integrating Factor,

$$[IF = e^{\int P(x)dx}.]$$

Bernoulli equation:

$$[y' + Py = Qy^n, \quad \text{where } v = y^{1-n}.]$$

Exact equation:

$$[Mdx + Ndy = 0, \quad \text{where } M_y = N_x.]$$

Orthogonal slope: If a curve has slope m , its orthogonal trajectory has slope $-1/m$.

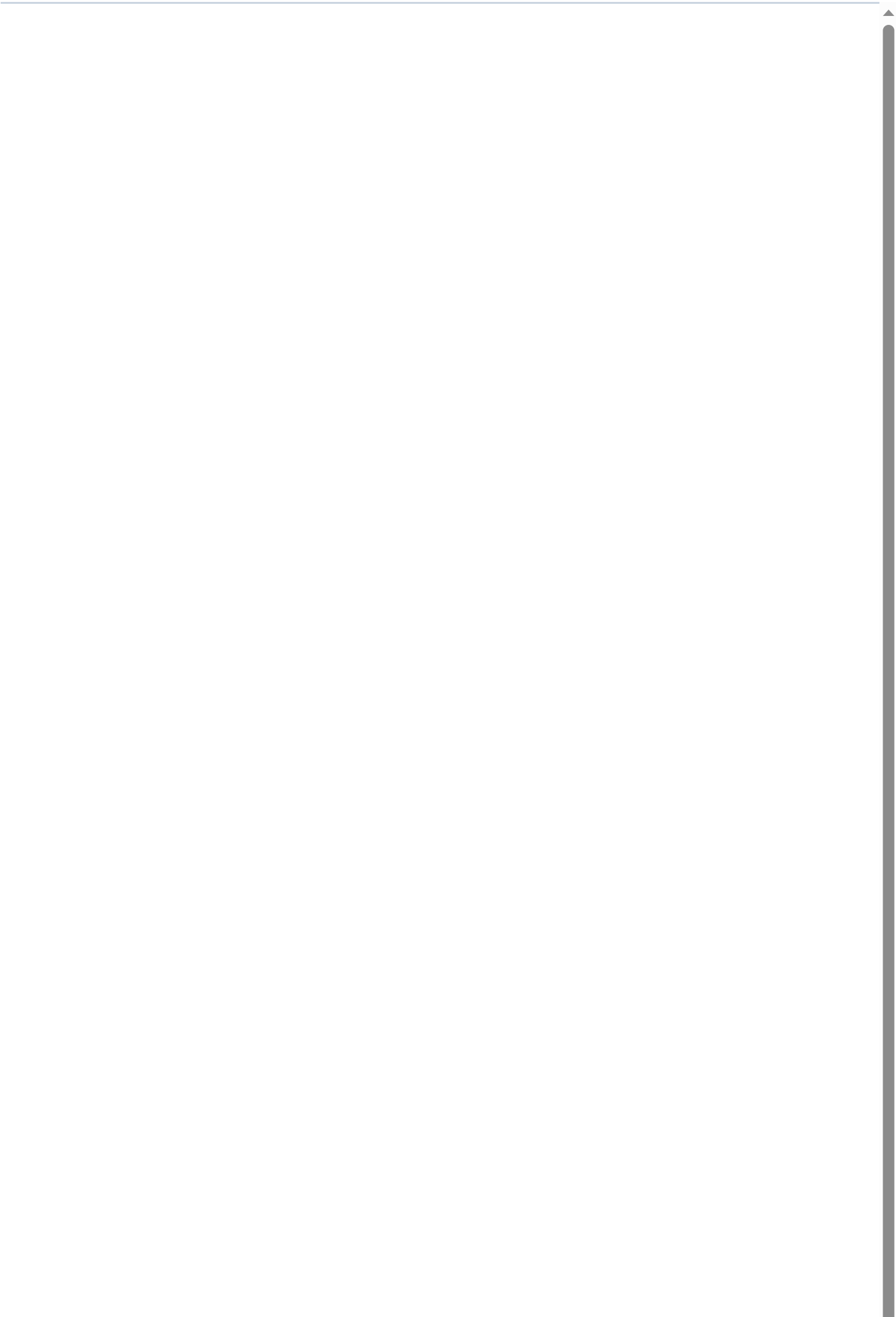
Newton's law of cooling:

$$[T - T_s = Ce^{-kt},]$$

where T_s is the surrounding temperature.

Clairaut equation:

$$[y = px + f(p).]$$



Module 4 – Higher-Order ODEs

Q7 – 20 marks

Q7(a) – 6 marks

Prompt: Solve

$$[y'' - 3y' + 2y = e^{3x}.]$$

Stepwise solution and marking

The auxiliary equation is

$$[m^2 - 3m + 2 = 0.]$$

[1 mark]

Factor:

$$[(m-1)(m-2) = 0,]$$

giving $m = 1, 2$. **[1 mark]**

Therefore, the complementary function is

$$[CF = C_1e^x + C_2e^{2x}.]$$

[1 mark]

For the particular integral,

$$[PI = \frac{e^{3x}}{3^2 - 3(3) + 2}.]$$

[1 mark]

Hence,

$$[PI = \frac{e^{3x}}{9 - 9 + 2} = \frac{1}{2}e^{3x}.]$$

[1 mark]

Thus,

$$[\boxed{y = C_1e^x + C_2e^{2x} + \frac{1}{2}e^{3x}}.]$$

[1 mark]

Q7(b) – 7 marks

Prompt: Solve

$$[y'' - y = x^2.]$$

Intuition: Solve the homogeneous equation and assume a polynomial particular integral.

Stepwise solution and marking

The auxiliary equation is

$$[m^2 - 1 = 0,]$$

giving $m = \pm 1$. **[1 mark]**

Therefore,

$$[CF = C_1e^x + C_2e^{-x}.]$$

[1 mark]

Assume

$$[y_p = Ax^2 + Bx + C.]$$

[1 mark]

Then

$$[y_p'' = 2A.]$$

Substitution gives

$$[2A - (Ax^2 + Bx + C) = x^2.]$$

[1 mark]

Comparing coefficients:

$$[-A = 1, \quad -B = 0, \quad 2A - C = 0.]$$

[1 mark]

Hence,

$$[A = -1, \quad B = 0, \quad C = -2,]$$

so

$$[y_p = -x^2 - 2.]$$

[1 mark]

Therefore,

$$[\boxed{y = C_1 e^x + C_2 e^{-x} - x^2 - 2}.]$$

[1 mark]

Q7 © 7 marks

Prompt: Using variation of parameters, solve

$$[y'' + y = \sec x.]$$

Stepwise solution and marking

The complementary solution is

$$[y_c = C_1 \cos x + C_2 \sin x.]$$

Thus choose

$$[y_1 = \cos x, \quad y_2 = \sin x.]$$

[1 mark]

The Wronskian is

$$[W = y_1 y_2' - y_1' y_2 = \cos^2 x + \sin^2 x = 1.]$$

[1 mark]

Variation of parameters gives

$$[y_p = -y_1 \int \frac{y_2 R}{W} dx + y_2 \int \frac{y_1 R}{W} dx,]$$

where $R = \sec x$. **[1 mark]**

Hence,

$$[y_p = -\cos x \int \sin x \sec x dx + \sin x \int \cos x \sec x dx.]$$

[1 mark]

Since

$$[\int \tan x dx = -\log |\cos x|, \quad \int 1 dx = x,]$$

[1 mark]

we get

$$[y_p = \cos x \log |\cos x| + x \sin x.]$$

[1 mark]

Therefore,

$$[\boxed{y = C_1 \cos x + C_2 \sin x + \cos x \log |\cos x| + x \sin x.}]$$

[1 mark]

OR

Q8 €” 20 marks

Q8(a) €” 6 marks

Prompt: Solve

$$[(D^2-1)(D^2+4)y=0, \quad D=\frac{d}{dx}.]$$

Stepwise solution and marking

The auxiliary equation is

$$[(m^2-1)(m^2+4)=0.]$$

[1 mark]

$$\text{From } m^2 - 1 = 0,$$

$$[m = \pm 1.]$$

[1 mark]

$$\text{From } m^2 + 4 = 0,$$

$$[m = \pm 2i.]$$

[1 mark]

The real-root contribution is

$$[C_1 e^x + C_2 e^{-x}.]$$

[1 mark]

The complex-root contribution is

$$[C_3 \cos 2x + C_4 \sin 2x.]$$

[1 mark]

Therefore,

$$[\boxed{y = C_1 e^x + C_2 e^{-x} + C_3 \cos 2x + C_4 \sin 2x.}]$$

[1 mark]

Q8(b) €” 7 marks

Prompt: Solve

$$[y'' + 3y' + 2y = \sin x.]$$

Stepwise solution and marking

The auxiliary equation is

$$[m^2 + 3m + 2 = 0 = (m+1)(m+2).]$$

[1 mark]

Hence,

$$[CF = C_1 e^{-x} + C_2 e^{-2x}.]$$

[1 mark]

Assume

$$[y_p = A \sin x + B \cos x.]$$

[1 mark]

Then

$$[y_p' = A \cos x - B \sin x, \quad y_p'' = -A \sin x - B \cos x.]$$

[1 mark]

Comparing coefficients after substitution gives

$$[A - 3B = 1, \quad 3A + B = 0.]$$

[1 mark]

Solving,

$$[A = \frac{1}{10}, \quad B = -\frac{3}{10}.]$$

[1 mark]

Therefore,

$$[\boxed{y = C_1 e^{-x} + C_2 e^{-2x} + \frac{1}{10} \sin x - \frac{3}{10} \cos x}.]$$

[1 mark]

Q8 © " 7 marks

Prompt: Solve the Cauchy–Euler equation

$$[x^2 y'' - xy' + y = x^2, \quad x > 0.]$$

Intuition: Substitute $t = \log x$, converting the Cauchy–Euler equation into a constant-coefficient equation.

Stepwise solution and marking

Put

$$[t = \log x.]$$

Then

$$[xy' = D_t y, \quad x^2 y'' = D_t(D_t - 1)y.]$$

[1 mark]

Therefore,

$$[[D_t(D_t - 1) - D_t + 1]y = e^{2t}.]$$

[1 mark]

Thus,

$$[(D_t - 1)^2 y = e^{2t}.]$$

[1 mark]

The repeated auxiliary root is $m = 1$. Hence,

$$[y_c = (C_1 + C_2 t)e^t.]$$

[1 mark]

Since $t = \log x$ and $e^t = x$,

$$[y_c = x(C_1 + C_2 \log x).]$$

[1 mark]

The particular integral is

$$[y_p = \frac{e^{2t}}{(2-1)^2} = e^{2t} = x^2.]$$

[1 mark]

Therefore,

$$[\boxed{y = x(C_1 + C_2 \log x) + x^2}.]$$

[1 mark]

Module 4 Formula Glossary

ODE: Ordinary Differential Equation.

$D = d/dx$: differential operator.

AE: Auxiliary Equation.

CF: Complementary Function.

PI: Particular Integral.

General solution:

$$[y = CF + PI.]$$

For $F(D)y = e^{ax}$:

$$[PI = \frac{e^{ax}}{F(a)} \quad \text{if } F(a) \neq 0.]$$

Variation of parameters:

$$[y_p = -y_1 \int \frac{y_2 R}{W} dx + y_2 \int \frac{y_1 R}{W} dx.]$$

Wronskian:

$$[W = y_1 y_2' - y_1' y_2.]$$

Cauchyâ€™Euler substitution:

$$[t = \log x, \quad xy' = D_t y, \quad x^2 y'' = D_t(D_t - 1)y.]$$

Module 5 – Linear Algebra

Q9 – 20 marks

Q9(a) – 6 marks

Prompt: Find the rank of

[A=

$$\begin{bmatrix} 1 & 2 & 3 & 2 & 4 & 6 & 1 & 1 & 1 \end{bmatrix}$$

]

using elementary row transformations.

Stepwise solution and marking

Start with

[A=

$$\begin{bmatrix} 1 & 2 & 3 & 2 & 4 & 6 & 1 & 1 & 1 \end{bmatrix}$$

.]

[1 mark]

Apply $R_2 \rightarrow R_2 - 2R_1$:

[

$$\begin{bmatrix} 1 & 2 & 3 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

.]

[1 mark]

Apply $R_3 \rightarrow R_3 - R_1$:

[

$$\begin{bmatrix} 1 & 2 & 3 & 0 & 0 & 0 & -1 & -2 \end{bmatrix}$$

.]

[1 mark]

Interchange the second and third rows:

[

$$\begin{bmatrix} 1 & 2 & 3 & 0 & -1 & -2 & 0 & 0 & 0 \end{bmatrix}$$

.]

[1 mark]

There are exactly two nonzero rows. **[1 mark]**

Therefore,

[$\boxed{\operatorname{rank}(A)=2}$.]

[1 mark]

Q9(b) € 7 marks

Prompt: Solve by the Gauss–Jordan method:

[$x+y+z=6$,]

[$2x-y+z=3$,]

[$x+2y-z=2$.]

Stepwise solution and marking

Write the augmented matrix:

[\left[

$$\begin{bmatrix} 1 & 1 & 1 & 6 & 2 & -1 & 1 & 3 & 1 & 2 & -1 & 2 \end{bmatrix}$$

\right].]

[1 mark]

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - R_1$:

[\left[

$$\begin{bmatrix} 1 & 1 & 1 & 6 & 0 & -3 & -1 & -9 & 0 & 1 & -2 & -4 \end{bmatrix}$$

\right].]

[1 mark]

Interchange R_2 and R_3 :

[\left[

$$\begin{bmatrix} 1 & 1 & 1 & 6 & 0 & 1 & -2 & -4 & 0 & -3 & -1 & -9 \end{bmatrix}$$

\right].]

[1 mark]

Apply $R_3 \rightarrow R_3 + 3R_2$:

[\left[

$$\begin{array}{cccc|cccc|c} 1 & 1 & 1 & 6 & 0 & 1 & -2 & -4 & 0 & 0 & -7 & -21 \end{array}$$

\right].]

[1 mark]

Divide the third row by -7 :

[$R_3 = [0 \ 0 \ 1 \mid \dots]$]

Thus $z = 3$. [1 mark]

Eliminating z and then y gives

[\left[

$$\begin{array}{cccc|cccc|c} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 2 & 0 & 0 & 1 & 3 \end{array}$$

\right].]

[1 mark]

Therefore,

[$\boxed{x=1, \quad y=2, \quad z=3}$.]

[1 mark]

Q9© " 7 marks

Prompt: Use five iterations of the Rayleigh power method to approximate the dominant eigenvalue and corresponding eigenvector of

[$A =$

$$\begin{bmatrix} 2 & 1 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{bmatrix}$$

,]

starting with

[$X_0 =$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

.]

Intuition: Repeated multiplication magnifies the component in the dominant eigenvector's direction. Normalize each result by its largest component.

Stepwise solution and marking

First iteration:

[Y_1=AX_O=

[2\1\1]

, \quad \lambda_1=2, \quad X_1=

[1\0.5\0.5]

.]

[1 mark]

Second iteration:

[Y_2=

[2.5\2\0.5]

, \quad \lambda_2=2.5, \quad X_2=

[1\0.8\0.2]

.]

[1 mark]

Third iteration:

[Y_3=

[2.8\2.6\0.2]

, \quad \lambda_3=2.8,]

[X_3\approx

[1\0.928571\0.071429]

.]

[1 mark]

Fourth iteration:

[Y_4\approx

[2.928571\2.857142\0.071429]

, \quad \lambda_4\approx 2.928571,]

[X_4\approx

[1\0.975610\0.024390]

.]

[1 mark]

Fifth iteration:

[Y_5\approx

[2.975610\2.951220\0.024390]

.]

[1 mark]

Therefore,

[\lambda_5\approx 2.975610, \quad X_5\approx

[1\0.991803\0.008197]

.]

[1 mark]

The sequence converges to

[\boxed{\lambda_{\max}=3}, \quad \boxed{X}\text{propto}

[1\1\0]

}.]

After five iterations, the requested approximation is

[\boxed{\lambda_{\max}}\approx 2.97561.]

[1 mark]

OR

Q10 " 20 marks

Q10(a) " 6 marks

Prompt: Determine the values of λ and μ for which the system

$$[x+y+z=3,]$$

$$[2x+3y+4z=9,]$$

$$[3x+4y+\lambda z=\mu]$$

has:

a unique solution,

infinitely many solutions,

no solution.

Stepwise solution and marking

Write the augmented matrix:

[\left[

$$\begin{array}{ccc|ccc|cc} 1 & 1 & 1 & 3 & 2 & 3 & 4 & 9 & 3 & 4 & \lambda & \mu \end{array}$$

\right].]

[1 mark]

Apply

$$[R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1:]$$

[\left[

$$\begin{array}{ccc|ccc|cc} 1 & 1 & 1 & 3 & 0 & 1 & 2 & 3 & 0 & 1 & \lambda - 3 & \mu - 9 \end{array}$$

\right].]

[1 mark]

Apply $R_3 \rightarrow R_3 - R_2$:

[\left[

$$\begin{array}{ccc|ccc|cc} 1 & 1 & 1 & 3 & 0 & 1 & 2 & 3 & 0 & 0 & \lambda - 5 & \mu - 12 \end{array}$$

\right].]

[1 mark]

If $\lambda \neq 5$, the third pivot is nonzero. Therefore, the solution is unique. **[1 mark]**

If $\lambda = 5$ and $\mu = 12$, the last row is all zero, so there are infinitely many solutions. **[1 mark]**

If $\lambda = 5$ and $\mu \neq 12$, the last row represents $0 = \mu - 12$, an inconsistency. Hence there is no solution. **[1 mark]**

Final classification:

[\boxed{\lambda \neq 5: \text{ unique solution}}]

[\boxed{\lambda = 5, \mu = 12: \text{ infinitely many solutions}}]

[\boxed{\lambda = 5, \mu \neq 12: \text{ no solution}}.]

Q10(b) € 7 marks

Prompt: Solve by the Gauss–elimination method:

$$[x+y+z=6,]$$

$$[2x+3y+z=11,]$$

$$[x-y+2z=5.]$$

Stepwise solution and marking

The augmented matrix is

[\left[

$$\begin{array}{ccc|ccc|ccc} 1 & 1 & 1 & 6 & 2 & 3 & 1 & 1 & 1 & -1 & 2 & 5 \end{array}$$

\right].]

[1 mark]

Apply $R_2 \rightarrow R_2 - 2R_1$:

$$[R_2=[0 \ 1 \ -1 \mid -1].]$$

[1 mark]

Apply $R_3 \rightarrow R_3 - R_1$:

$$[R_3=[0 \ -2 \ 1 \mid -1].]$$

[1 mark]

Apply $R_3 \rightarrow R_3 + 2R_2$:

$$[R_3=[0 \ 0 \ -1 \mid -3].]$$

[1 mark]

Therefore,

$$[-z=-3,\text{quad } z=3.]$$

[1 mark]

From $y - z = -1$,

$$[y-3=-1,\text{quad } y=2.]$$

From $x + y + z = 6$,

$$[x=1.]$$

[1 mark]

Hence,

$$[\boxed{x=1,\text{quad } y=2,\text{quad } z=3}.]$$

[1 mark]

Q10© € 7 marks

Prompt: Use the Gauss–Seidel method to solve

$$[10x+y+z=12,]$$

$$[2x+10y+z=13,]$$

$$[2x+2y+10z=14,]$$

starting from $(0, 0, 0)$ and carrying out four iterations.

Intuition: Rearrange each equation to isolate its dominant variable, then immediately use every newly calculated value.

Stepwise solution and marking

Iteration formulas:

$$[x = \frac{12 - y - z}{10},]$$

$$[y = \frac{13 - 2x - z}{10},]$$

$$[z = \frac{14 - 2x - 2y}{10}.]$$

[1 mark]

Starting with $(x_0, y_0, z_0) = (0, 0, 0)$:

$$[x_1 = 1.2, \text{quad } y_1 = 1.06, \text{quad } z_1 = 0.948.]$$

[1 mark]

Second iteration:

$$[x_2 = \frac{12 - 1.06 - 0.948}{10} = 0.9992,]$$

$$[y_2 = \frac{13 - 2(0.9992) - 0.948}{10} = 1.00536,]$$

$$[z_2 = \frac{14 - 2(0.9992) - 2(1.00536)}{10} = 0.999088.]$$

[1 mark]

Third iteration:

$$[x_3 = 0.9995552,]$$

$$[y_3 = 1.00018016,]$$

$$[z_3 = 1.000052928.]$$

[1 mark]

Fourth iteration:

$$[x_4 = 0.9999766912,]$$

$$[y_4 = 0.99999936896,]$$

$$[z_4 = 1.000004787968.]$$

[1 mark]

To four decimal places,

$$[(x, y, z) \approx (1.0000, 1.0000, 1.0000).]$$

[1 mark]

Direct verification:

$$[10(1) + 1 + 1 = 12,]$$

$$[2(1) + 10(1) + 1 = 13,]$$

$$[2(1) + 2(1) + 10(1) = 14.]$$

Therefore,

$$[\boxed{x=1, \text{quad } y=1, \text{quad } z=1}.]$$

[1 mark]

Module 5 Formula Glossary

Rank: Number of nonzero rows in a row-echelon form.

Augmented matrix: Matrix obtained by adjoining the constant column to the coefficient matrix.

Consistency conditions:

Unique solution:

[\operatorname{rank} (A)

$\operatorname{rank}([A|B]) = n.$

Infinitely many solutions:

[rank] (A)

$\text{rank}([A|B]) < n.$]

No solution:

[$\text{rank}(A) \neq \text{rank}([A|B])$.]

Gauss elimination: Converts a system to upper-triangular form, followed by back-substitution.

Gauss-Jordan method: Reduces the augmented matrix to reduced row-echelon form.

Gauss-Seidel method: Iterative method that immediately uses newly computed variable values.

Power method: Iterative procedure for finding the dominant eigenvalue and eigenvector.

Dominant eigenvalue: Eigenvalue having the largest absolute magnitude.

Eigenvalue and eigenvector:

[$AX = \lambda X$.]

Arithmetic Verification Summary

Every full question totals

[$6+7+7=\boxed{20\text{ marks}}.$]

Five selected full questions total

[$5(20)=\boxed{100\text{ marks}}.$]

The following key results were independently substituted back or numerically checked:

Q3(c): minimum value 7 at $(2, -3)$.

Q5(a): $y^{-1} = x + 1 + Ce^x$ satisfies the transformed linear equation.

Q6(b): temperature excesses follow 60, 30, 15.

Q7(a), Q7(b), Q8(b), and Q8(c): particular integrals satisfy their respective ODEs.

Q9(b) and Q10(b): $(1, 2, 3)$ satisfies all three equations.

Q9(c): exact dominant eigenpair is $\lambda = 3, X \propto (1, 1, 0)^T$.

Q10(c): $(1, 1, 1)$ satisfies the system exactly.